

Exam 1 - MATH-132B-01

Instructions: You have 110 minutes to take this exam. You are permitted your signed prep work journal and access to RStudio on a classroom computer during the exam. No other notes or electronic devices of any kind are allowed on this exam. Failure to comply may result in earning a 0 on any portion of this exam and/or being reported to the university.

Multiple Choice: There are 10 multiple choice questions, each worth 4 points. Select one answer in each question by circling the letter corresponding to your selected answer. There is only one correct answer for each question. You will not be penalized for incorrect answers.

For problems 1-2, consider the following experiment. A fair six-sided die is rolled once and the result is recorded.

1. Which of the following events is simple?

- A. Rolling an even number
- B. Rolling a number greater than 2
- C. Rolling a 6
- D. Rolling a prime number

2. Let E be the event a prime number is rolled. Which of the following events are disjoint from E ?

- A. Rolling a 6
- B. Rolling an even number
- C. Rolling a 3
- D. Rolling an odd number

For problems 3-5, consider the following experiment. A bag contains 30 marbles; 23 of them have stripes and 15 of them have spots. All marbles in the bag have at least one pattern. A marble is drawn from the bag at random.

3. What is the probability a striped marble is drawn?

- A. $\frac{23}{30}$
- B. $\frac{15}{30}$
- C. $\frac{8}{30}$
- D. $\frac{7}{30}$

4. What is the probability a marble is drawn that is NOT spotted?

- A. $\frac{3}{5}$
- B. $\frac{1}{3}$
- C. $\frac{1}{5}$
- D. $\frac{1}{2}$

5. What is the probability a marble is drawn that is spotted AND striped?

- A. $\frac{1}{2}$
- B. $\frac{4}{15}$
- C. $\frac{1}{6}$
- D. $\frac{19}{15}$

For problems, consider the following experiment. A pharmaceutical company is testing their new drug for plaque psoriasis. They are able to recruit 60 female and 40 male afflicted individuals for the trial. The drug is perscribed to 30 randomly selected females and 20 randomly selected males. A placebo is assigned to the remaining participants. To measure the effects of the drug, they ask patients to return every 2 weeks over the next 3 months. The percentage of the body afflicted is measured at each visit.

6. What sampling method was used in this experiment?

- A. Random B. Stratified C. Systematic D. Cluster

7. Which of the following best describe each measurement taken from a patient?

- A. Statistic B. Parameter C. Datum D. Sample

8. What type of data is being recorded in this experiment?

- A. Nominal B. Ordinal C. Interval D. Ratio

9. What is the sample size of this experiment?

- A. 100 B. 60 C. 40 D. 50%

10. Which of the following best describe the nature of the study?

- A. Restrospective B. Cross-Sectional C. Prospective D. Nonparametric

Free Response: There are free-response questions worth 70 points total. For each problem, record your solution in the space provided. All answers **must have justification** in order to receive credit. If you have used an R command to compute your solution, refer to the data set as `df` and **write the code you used as justification** for your answer. If you need additional room, you may attach a piece of the provided colored scratch paper. In this case, please write your name on each sheet you attach and clearly indicate which question the attached work corresponds to.

11. [13 points] A scientist is studying the effects of a pesticides on earwigs. They have a sample of 200 earwigs, 140 of which are European earwigs and 60 of which are striped earwigs. They plan to expose a randomly-selected group of the earwigs to the pesticide.

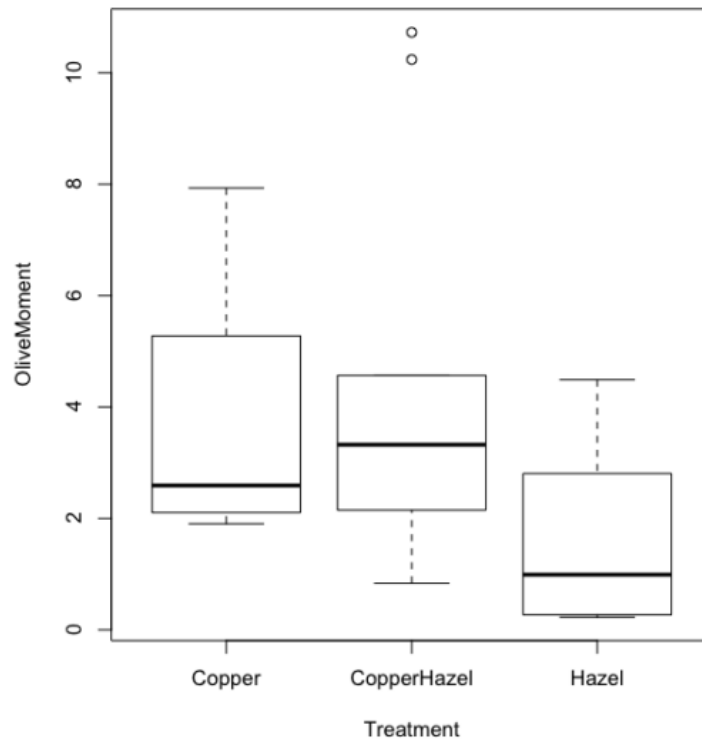
(a) Describe a sampling technique that could potentially introduce a nonrandom sampling error in this study.

(b) Describe a way the scientist could design a stratified sample for this experiment. Then, give an example of how a nonsampling error could be obtained from this study.

(c) Describe a way the scientist could design a clustered sample for this experiment. Then, give an example of how a sampling error could be obtained from this study.

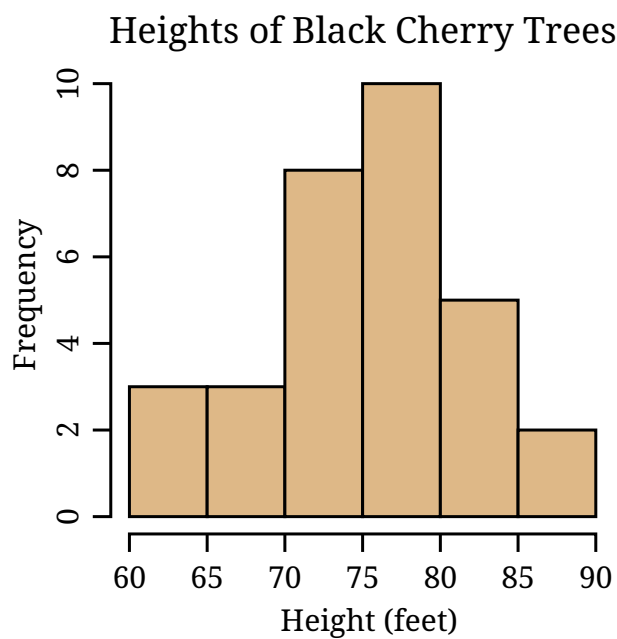
(d) Describe the data the scientist is recording by listing its type, level of measurement, and whether it is discrete, continuous, or neither.

12. [15 points] The boxplots given below each show olive moments from rat lung tissue samples. Each boxplot corresponds to a different treatment.



- (a) Identify any outlier(s) in each data set.
- (b) Use the image to estimate the 5 number summary for the "Copper" data set.
- (c) Is the "Copper" data set or the "Hazel" data set more normal? Justify your answers.
- (d) Which data set has the widest IQR?
- (e) Which data set has the widest range?

13. [9 points] The histogram below shows the measured heights of black cherry trees.



(a) How many cherry trees were sampled?

(b) How many trees are less than 75 feet tall?

(c) How many cherry trees are between 70 and 85 feet tall?

(d) Use the histogram to describe the shape of the data.

14. [20 points] The heights of 11 patients (in inches) are given below.

62 64 69 70 70 71 72 73 74 75 77

(a) Compute the mean height.

(b) Find the median height.

(c) Determine the inner-quartile range of the data.

(d) Determine the range of the data.

(e) Write the 5-number summary for the data.

(f) Find the 35th percentile.

(g) Compute the standard deviation from the sample.

(h) Does the data have any outliers? Why or why not?

(i) Sketch a boxplot for the data in the space below. Clearly label your number line with all relevant quantities.

15. [13 points] The scores for 11 students on a biostatistics exam are given in the table below.

40 55 61 65 68 70 72 74 80 84 85

Plot a histogram with 5 bins in the axes provided below. Clearly label your axes and the endpoints of each bin.

